

Faculty of Engineering & Technology					
Ramaiah University of Applied Sciences					
Department	Computer Science and Engineering		Programme	B. Tech. in CSE,ISE	
Semester/Batch	6 th /2023, 2022, 2021				
Course Code	AID201A		Course Title	Artificial Intelligence/Principles of AI	
Course Leader	Jishmi Jos Choondal				
Assignment					
Name of Student				Register No	
Sections		Marking Scheme	Max Marks	First Examiner Marks	Second Examiner Marks
A		Data pre-processing and exploration	4		
		Feature engineering and selection	4		
		Model selection and justification	4		
		Model training and evaluation	4		
		IEEE format documentation	4		
		Max Marks	20		
Total Assignment Marks			20		

Course Marks Tabulation				
Component- CET B Assignment	First Examiner	Remarks	Second Examiner	Remarks
A				
Marks (out of 20)				
Signature of First Examiner		Signature of Second Examiner		

Please note:

1. Documental evidence for all the components/parts of the assessment such as the reports, photographs, laboratory exam / tool tests are required to be attached to the assignment report in a proper order.

2. The First Examiner is required to mark the comments in RED ink and the Second Examiner's comments should be in GREEN ink.
3. The marks for all the questions of the assignment have to be written only in the **Component – CET B: Assignment** table.
4. If the variation between the marks awarded by the first examiner and the second examiner lies within ± 3 marks, then the marks allotted by the first examiner is considered to be final. If the variation is more than ± 3 marks then both the examiners should resolve the issue in consultation with the Chairman BoE.

Assignment

Instructions to students:

1. The assignment consists of 1 question.
2. Maximum marks is 20.
3. The assignment has to be neatly word processed as per the prescribed format.
4. The maximum number of pages should be restricted to 6.
5. Restrict your report to a maximum of 6 pages.
6. The assignment must be submitted to the subject leader via msruas portal.
7. **Submission Date: 18th May 2026**
8. **Submission after the due date is not permitted.**
9. **IMPORTANT:** It is essential that all the sources used in preparation of the assignment must be suitably referenced in the text.
10. Marks will be awarded only to the sections and subsections clearly indicated as per the problem statement/exercise/question

PART A

20 Marks

In this Assignment, students are required to design and develop a project on Machine Learning. The project can be focused on either a regression or classification problem, where you will be predicting a target variable based on various features. You are free to use any appropriate machine learning algorithm for this problem, but you must justify your choice of algorithm. You must use a publicly available dataset for this assignment.

You must perform the following tasks:

- Data exploration and pre-processing: Perform data cleaning, visualization, and pre-processing.
- Feature engineering: Create new features if needed and perform feature selection if necessary.
- Model selection: Choose an appropriate algorithm for the problem and justify your choice.
- Model training and evaluation: Train the model on the dataset and evaluate its performance using appropriate metrics.

You must submit the following:

The report shall be prepared in IEEE format and must comprehensively summarize the abstract, methodology, results, conclusions, and references. Additionally, the marking scheme should be presented on the front page of the report.

Shipping Delay Prediction using PySpark

ABSTRACT:

Predictability in supply chain logistics is crucial for maintaining customer satisfaction in e-commerce. This project focuses on a binary classification problem to predict whether an e-commerce shipment will arrive on time or experience delays. Utilizing Apache Spark (PySpark) and its Machine Learning library (MLlib), a predictive model was developed using a Logistic Regression algorithm. The pipeline encompasses data pre-processing, feature engineering via String Indexing and Vector Assembling, and model evaluation based on various package and transit factors.

1. INTRODUCTION

The rapid expansion of the global e-commerce sector has fundamentally transformed consumer expectations, placing immense pressure on logistical networks to deliver packages within strict, expedited timeframes. In today's highly competitive market, the efficiency of supply chain management is a critical determinant of business success. Delayed shipments cascade into a series of negative consequences, including increased operational routing costs, overwhelmed customer support centers, and a significant drop in customer ratings and long-term retention. Managing these complex logistics involves navigating a multitude of dynamic variables, from warehouse operations and diverse transit modes (such as flight, ship, or road) to handling packages of varying weights and financial importance. Traditional, rule-based tracking systems often struggle to process the sheer volume of transit data or identify the subtle, non-linear patterns that lead to logistical bottlenecks.

To address this challenge, leveraging machine learning offers a proactive, data-driven solution. By analyzing massive datasets of historical shipping records, predictive models can identify the underlying factors that consistently contribute to late deliveries. This project utilizes Apache Spark's powerful big data framework, PySpark, to build a robust Logistic Regression classification model. The primary objective is to calculate the precise probability of a shipping delay based on specific transit and product factors, ultimately empowering businesses to optimize their inventory management, mitigate transit risks, and maintain transparent, effective communication with their customers.

2. DATASET DESCRIPTION

The model is trained on a publicly available dataset containing historical e-commerce shipping records. The primary objective is to predict a target variable indicating a delayed or on-time delivery based on several features.

Key features analyzed in the dataset include:

- **Warehouse_block:** The specific warehouse facility where the item originated.
- **Mode_of_Shipment:** The method of transit (Flight, Ship, or Road).
- **Product_importance:** Categorized as low, medium, or high.
- **Cost_of_the_Product:** The financial value of the item.
- **Discount_offered:** The percentage of discount applied to the purchase.
- **Weight_in_gms:** The physical weight of the package.
- **label:** The target variable (formerly **Reached_on_Time_Y_N**), representing if the package was on time (1.0) or delayed (0.0).

3. DATA EXPLORATION AND PRE-PROCESSING

Data pre-processing is a critical step to ensure the dataset is compatible with PySpark's MLlib requirements. The raw data was processed using the following steps:

- **Column Formatting:** Spark processes data more efficiently when column names avoid certain characters. All periods (".") in the column headers were replaced with underscores ("_").
- **Data Cleaning:** Null values and inconsistencies were handled by dropping rows with missing data. Furthermore, the "ID" column was dropped as it serves only as a unique identifier and provides no predictive value.
- **Target Variable Renaming:** The original target column, **Reached_on_Time_Y_N**, was renamed to **label** to align with PySpark MLlib's standard naming conventions for classification targets.

4. FEATURE ENGINEERING

To prepare the cleaned data for model training, specific feature engineering techniques were applied to convert categorical text data into a machine-readable numerical format.

- **String Indexing:** Categorical variables such as `Warehouse_block`, `Mode_of_Shipment`, `Product_importance`, and `Gender` were transformed using PySpark's `StringIndexer`. This function assigns a numerical index to each unique string value.
- **Vector Assembling:** Machine learning models in PySpark require features to be passed as a single unified vector. The `VectorAssembler` function was utilized to concatenate the newly created numerical indices alongside the existing continuous numeric columns (`Customer_care_calls`, `Customer_rating`, `Cost_of_the_Product`, `Prior_purchases`, `Discount_offered`, `Weight_in_gms`) into a single output column named `features`.

5. MODEL SELECTION

A Logistic Regression classification algorithm was selected to calculate the probability of a shipping delay.

Logistic Regression is highly appropriate for this problem because it is specifically designed for binary classification tasks (predicting between two distinct outcomes: delayed or on time). Furthermore, it outputs probability scores, allowing businesses to understand not just *if* a package will be delayed, but the *likelihood* of that delay. Within the PySpark ecosystem, Logistic Regression is highly scalable and computationally efficient when processing large tabular datasets.

6. MODEL TRAINING AND EVALUATION

The model training was streamlined using PySpark's `Pipeline` API. The pipeline integrated the `StringIndexer` stages, the `VectorAssembler`, and the `LogisticRegression` estimator into a single workflow, ensuring that the same transformations applied to the training data would be consistently applied to any new testing data.

To evaluate the model's practical application, it was tested against two extreme simulated scenarios:

- **Best Case Scenario:** A high-importance product shipped via Flight, with high customer

ratings and a high discount. The model correctly predicted **1.0** (On Time).

- **Worst Case Scenario:** A low-importance product shipped via Ship, with poor customer ratings, high customer care calls, and low discount. The model accurately predicted **0.0** (Delayed).

Some of the values I used for training the model are given below:

```
Shipping_prediction-main > shipping.csv
1 ID,Warehouse block,Mode_of_Shipment,Customer_care_calls,Customer_rating,Cost_of_the_Product,Prior_purchases,
2 1,D,Flight,4,2,177,3,low,F,44,1233,1
3 2,F,Flight,4,5,216,2,low,M,59,3088,1
4 3,A,Flight,2,2,183,4,low,M,48,3374,1
5 4,B,Flight,3,3,176,4,medium,M,10,1177,1
6 5,C,Flight,2,2,184,3,medium,F,46,2484,1
7 6,F,Flight,3,1,162,3,medium,F,12,1417,1
8 7,D,Flight,3,4,250,3,low,F,3,2371,1
9 8,F,Flight,4,1,233,2,low,F,48,2804,1
10 9,A,Flight,3,4,150,3,low,F,11,1861,1
11 10,B,Flight,3,2,164,3,medium,F,29,1187,1
12 11,C,Flight,3,4,189,2,medium,M,12,2888,1
13 12,F,Flight,4,5,232,3,medium,F,32,3253,1
14 13,D,Flight,3,5,198,3,medium,F,1,3667,1
15 14,F,Flight,4,4,275,3,high,M,29,2602,1
16 15,A,Flight,4,3,152,3,low,M,43,1009,1
17 16,B,Flight,4,3,227,3,low,F,45,2707,1
18 17,C,Flight,3,4,143,2,medium,F,6,1194,1
19 18,F,Ship,5,5,227,3,medium,M,36,3952,1
20 19,D,Ship,5,5,239,3,high,M,18,2495,1
21 20,F,Ship,4,5,145,3,medium,M,45,1059,1
22 21,A,Ship,3,3,161,2,medium,F,38,1521,1
23 22,B,Ship,3,1,232,4,medium,F,51,2899,1
24 23,C,Ship,2,5,156,2,low,M,2,1750,1
25 24,F,Ship,4,3,211,3,high,M,12,3922,1
26 25,D,Ship,4,5,251,2,medium,F,28,3561,1
27 26,F,Ship,3,1,225,4,low,M,29,3496,1
28 27,A,Ship,4,1,172,3,high,F,24,1066,1
29 28,B,Ship,5,1,162,3,medium,M,31,1435,1
30 29,C,Ship,2,3,234,4,low,M,44,3134,1
31 30,F,Ship,5,4,183,2,low,F,36,3819,1
32 31,D,Ship,3,4,266,2,medium,F,38,2741,1
33 32,F,Ship,5,3,257,2,low,F,61,3341,1
34 33,A,Ship,3,1,223,3,low,F,22,3795,1
35 34,B,Ship,4,2,234,3,medium,M,4,3261,1
36 35,C,Ship,4,2,227,4,medium,F,22,2460,1
37 36,F,Ship,5,4,149,3,medium,F,44,1811,1
38 37,D,Ship,3,5,137,4,high,F,62,1477,1
39 38,F,Ship,2,4,181,6,medium,F,16,3605,1
40 39,A,Ship,5,1,215,4,low,F,56,3905,1
```



```

# Convert text columns to numbers
categorical_cols = ["Warehouse_block", "Mode_of_Shipment", "Product_importance", "Gender"]
indexers = [StringIndexer(inputCol=c, outputCol=c+"_idx") for c in categorical_cols]

# Combine all features into one vector
numeric_cols = ["Customer_care_calls", "Customer_rating", "Cost_of_the_Product",
                "Prior_purchases", "Discount_offered", "Weight_in_gms"]
feature_cols = numeric_cols + [c+"_idx" for c in categorical_cols]

assembler = VectorAssembler(inputCols=feature_cols, outputCol="features")

print("Features ready!")

```

... Features ready!

```

# Define Model and Pipeline
lr = LogisticRegression(featuresCol="features", labelCol="label")
pipeline = Pipeline(stages=indexers + [assembler, lr])

print("Training model... (This takes a few seconds)")
model = pipeline.fit(df)
print("Model trained successfully!")

```

... Training model... (This takes a few seconds)
Model trained successfully!

```

# BEST CASE SCENARIO
new_data = spark.createDataFrame([
    ("F", "Flight", "high", "M", 2, 5, 150, 5, 65, 1200)
], ["Warehouse_block", "Mode_of_Shipment", "Product_importance", "Gender",
    "Customer_care_calls", "Customer_rating", "Cost_of_the_Product",
    "Prior_purchases", "Discount_offered", "Weight_in_gms"])

# Predict
result = model.transform(new_data)
final_output = result.select("prediction")

print("Prediction: 1.0 = On Time | 0.0 = Delayed\n")

display(final_output.toPandas())

```

... Prediction: 1.0 = On Time | 0.0 = Delayed

prediction

0	1.0
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Worst Case Scenario

```
new_data = spark.createDataFrame([
    ("A", "Ship", "low", "M", 7, 1, 300, 2, 1, 7500)
], ["Warehouse_block", "Mode_of_Shipment", "Product_importance", "Gender",
    "Customer_care_calls", "Customer_rating", "Cost_of_the_Product",
    "Prior_purchases", "Discount_offered", "Weight_in_gms"])

# Run the prediction
result = model.transform(new_data)
final_output = result.select("prediction")

print("Prediction: 1.0 = On Time | 0.0 = Delayed\n")

display(final_output.toPandas())
```

Prediction: 1.0 = On Time | 0.0 = Delayed

prediction

0	0.0
---	-----

7. CONCLUSION

This project successfully demonstrates the application of Apache Spark and Logistic Regression to solve a real-world supply chain classification problem. By implementing a robust pipeline of data cleaning, string indexing, and feature assembling, the model can process diverse shipment parameters to output actionable delay predictions.

8. REFERENCES

- [1] Apache Spark Documentation. "MLlib: Main Guide." *spark.apache.org*.
- [2] Public E-commerce Shipping Dataset. (Source: Kaggle).